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Browse by career focus, then open a case for methods, evidence and engineering judgement.

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01 Prusa eight-station assembly

Industrial engineering · Manufacturing



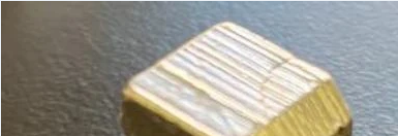
02 Measurement, quality & reliability

Quality · Manufacturing



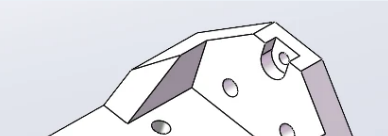
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04 Five processes & dimensional variation

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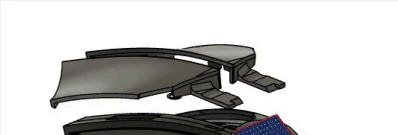
05 Part design for manufacture

Manufacturing · Quality



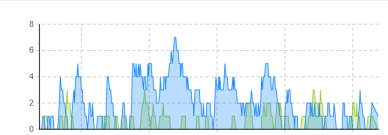
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Manufacturing · Industrial Engineering

Prusa MK4/MK4S: Eight-Station Assembly & Standard Work

Team assembly project · Time Studies · DFA · Line Balancing · Work Instructions

Focused on the Y-axis subassembly, recording assembly times, locating and fastening issues, and translating workshop observations into process improvement and standard work recommendations.

My contribution

I mainly carried out time studies, identified assembly issues and analysed design for assembly for the Y-axis subassembly. I also contributed to the work instructions and technical report.

Project outputs

An assembly time baseline, a Y-axis issue log, workstation improvement recommendations and standard work materials.

8

Assembly trial workstations

52.65 min

Measured baseline for total line assembly time

Y-axis

My main analysis focus

Issues & actions

The Y-axis assembly involves consecutive operations on guide rods, bearings, belts and belt tensioning. Repeated alignment, awkward locating or an unclear fastening sequence can increase both task time and assembly variation.

I broke down the actual operations and recorded process handoffs, then examined locating, tolerances, fastening and tool setup to understand repeated adjustments. I also discussed recommendations such as task reallocation, locating fixtures, assembly lead-in features and torque calibration.

Outputs developed

A total line assembly time baseline, a Y-axis issue log, workstation improvement recommendations, and work instructions and a technical report to which I contributed.

From assembly actions To engineering causes

Observation area	Analysis focus
Guide rods & bearings	Locating, alignment & fit
Belts & tensioning	Routing, tension & repeated adjustment
Fastened joints	Operation sequence, tools & torque
Workstation handoffs	Task load, waiting & handovers

Compiled from existing project records; this does not specify how tasks were assigned to the eight numbered workstations.

PROCESS & ENGINEERING JUDGEMENT

Assembly Tasks, Handoffs & Standard Work

Using the Y-axis subassembly as a specific case, I connected operational issues, process dependencies and improvement recommendations from the assembly trial. The focus was to explain where time was spent and how subsequent improvements could be verified.

01 Break down the actual assembly tasks

The Y-axis subassembly involves installing guide rods and bearings, routing and tensioning the belt, locating parts and tightening screws. I timed these consecutive tasks and recorded issues, noting where repeated alignment, readjustment and tool changes occurred. I considered assembly tasks alongside workstation handoffs to distinguish waiting, rework and direct operation time.

02 Trace observations to assembly design issues

Difficult locating may relate to lead-in features, fit tolerances or access space. Restricted movement after fastening calls for a review of the locating and fastening sequence. My analysis covered locating, tolerances, fastening and tool setup, informing discussion of locating aids, assembly lead-ins, dedicated tool setups, torque requirements and workstation task reallocation.

03 Turn observations into repeatable work requirements

The work instructions and technical report I helped prepare drew on workshop records and improvement discussions. In my review, I treated standard work as a combination of operation sequences, key inspection points and workstation handover requirements. The 52.65-minute figure is the baseline total line assembly time for this trial. Any improvement still needs to be verified by remeasuring workstation times, waiting, rework and assembly consistency.

Analysis area	Engineering considerations
Guide rods, bearings & locating	Fit, lead-ins, repeated alignment & smooth movement

Analysis area	Engineering considerations
Belts, tensioning & fastening	Assembly sequence, repeated adjustment, tools & torque requirements
Workstations & standard work	Task load, waiting and handoffs, inspection points & consistency across operators

Project basis: existing Prusa MK4/MK4S assembly records and experience preparing work instructions and a technical report. The original time-study sheet is not attached.

Engineering judgement

The 52.65-minute figure describes the total line assembly time measured in this trial. Assessing sustained production capacity also requires the bottleneck workstation cycle time, effective working time, staffing, waiting and rework data.

Improvement recommendations need to be retested with the same staffing, materials and tools to establish whether total time, workstation loading and assembly consistency improve together. The available materials contain no verifiable post-improvement percentage.

Improvement area	Recommendation	Follow-up checks
Locating & assembly sequence	Use locating aids and specify locating before tightening	Number of readjustments & smooth movement
Tools & fastening	Standardise tool setup and define critical fastening requirements	Fastening consistency, rework & omitted assembly steps
Workstation loading	Consider task reallocation in light of process dependencies	Changes in workstation times & waiting
Standard work	Document operations, inspection points and handover requirements in work instructions	Consistency across operators

Data and scope

A course assembly trial. The 52.65-minute figure is the measured baseline for total assembly time. Task reallocation, fixtures and fastening optimisation are recommendations whose effects require retesting under the same conditions.

Source: Prusa assembly project experience and standard work documentation. The original time-study sheet is not included in the reports attached here.

3D-Printed Hinges: Measurement Systems, Capability & Reliability

Individual analysis report

Minitab · Gage R&R · Process Capability · FMEA · Weibull

First assess whether the measurement data are trustworthy, then analyse dimensional capability and failure life. More consistent dimensions and greater durability are treated as separate questions requiring separate validation.

My contribution

My individual report covers inspection definitions, crossed Gage R&R, dimensional capability analysis, process FMEA and a fishbone diagram, plus a life comparison of the original and new hinges.

Project outputs

MSA for three dimensions, process capability analysis, FMEA / fishbone analysis and a life comparison of the original and new batches.

10 × 3 × 3

Measurement design for each dimension

9.81%

Thickness %GRR

20 parts

Total samples in original and new life tests

Validate the Measurement System Before the Process

For 10 parts, 3 operators each took 3 repeated measurements. Minitab crossed Gage R&R was used to analyse thickness, width and length. The thickness measurement system was usable; width and length had greater measurement error, so locating, measurement methods and repeatability needed priority attention.

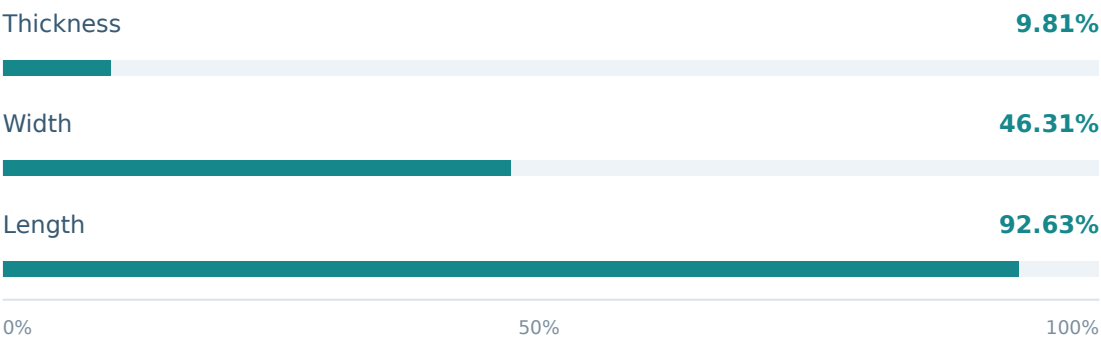
Dimension	%GRR	ndc	Report assessment
Thickness	9.81%	14	Measurement system usable
Width	46.31%	2	Prioritise measurement improvements
Length	92.63%	1	Measurement system unacceptable

Capability Analysis & Risk Prioritisation

Thickness $Ppk \approx 0.35$ and $Cpk \approx 0.27$ indicated insufficient process capability in the sample. The printing process map, FMEA and fishbone diagram supported discussion of improvements to slicing parameters, material drying, temperature control, locating and work instructions.

%GRR by dimension

% of study variation · 10 parts × 3 operators × 3 repeats



Redrawn using data from PDF pp. 10–21 of the original report. See the explanation and table on the left for the corresponding measurement system assessments.

PROCESS & ENGINEERING JUDGEMENT

From Measurement Confidence to Capability & Life

The report analyses measurement systems, dimensional capability, process risk and bending life separately, then considers how they jointly affect part quality.

01 Check whether the data support the assessment

Each dimension used a crossed design with 10 parts, 3 operators and 3 repeated measurements per operator. Gage R&R separates repeatability, reproducibility and part-to-part variation; ndc helps assess the ability to distinguish between parts. The thickness results support further discussion. Width and length first require improvements to locating, contact method and consistency of repeated measurements.

02 Link capability metrics to the manufacturing process

Thickness Ppk of approximately 0.35 and Cpk of approximately 0.27 indicated insufficient process capability in this sample. The report then used a process map, FMEA and fishbone diagram to organise factors including slicing, material drying, temperature control and measurement, linking failure modes to potential causes and preventive checks. Capability metrics help identify problems but cannot, on their own, establish which printing parameter caused the deviation.

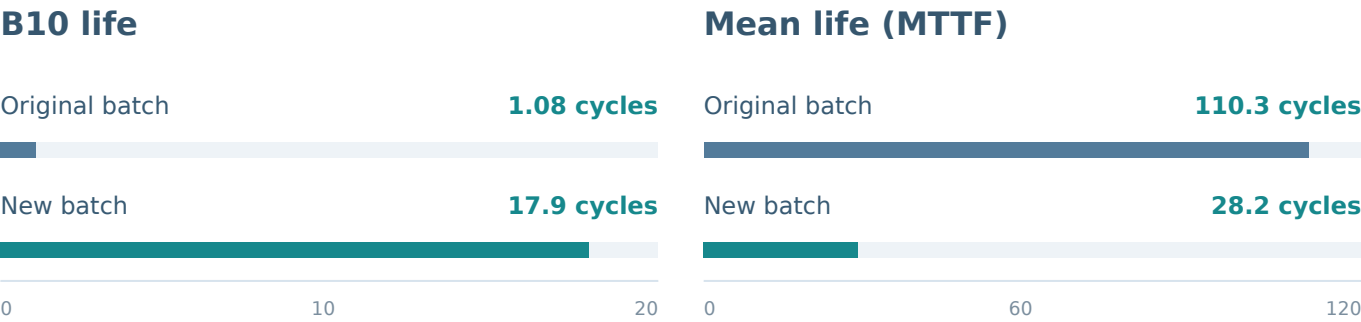
03 Validate dimensional quality and durability separately

Bending data from the original and new hinge groups, with 10 parts each, were compared using Weibull models. The new group had a higher B10 but a lower estimated mean life, showing that early failures and the long-life tail can lead to different conclusions. Evaluating a new design requires consideration of dimensional centring, variation, failure distributions and sample size. Improvement in one metric does not establish better overall reliability.

Analysis level	Question addressed
MSA / Gage R&R	Do the observed differences come from the parts or the measurement process?

Analysis level	Question addressed
Process Capability / FMEA	Do dimensions meet requirements, and which process factors should be reviewed first?
Weibull / B10 / MTTF	Do early failures, the life distribution and the mean support the same conclusion?

Original report: Probability, Reliability and Quality / PDF pp. 10–21, 29–43, 45–46 and 82–86.

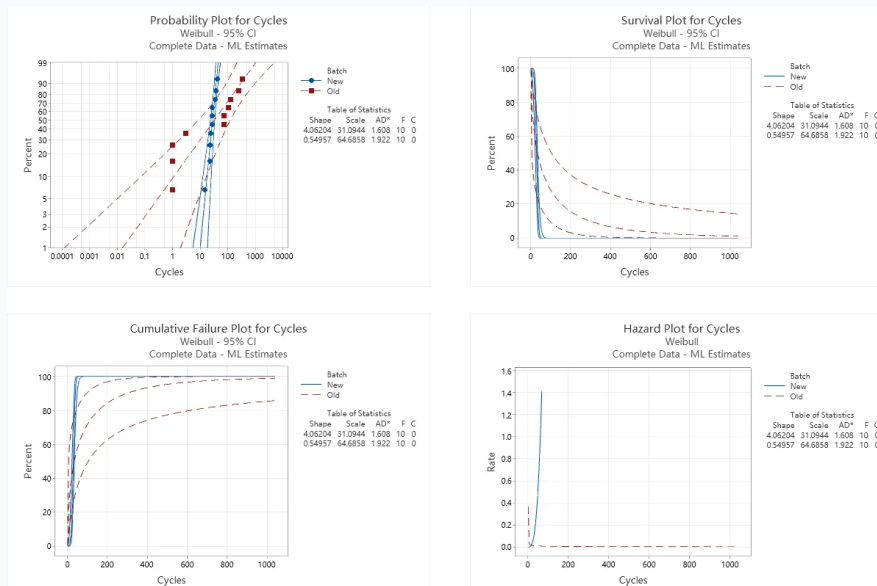


Weibull point estimates, 10 parts per batch; the two charts use separate horizontal scales. Small samples lead to substantial interval uncertainty; confidence intervals are not shown. Source: original report, PDF pp. 82–86.

Original vs New Batch Life

The life data, with 10 parts per group, showed different failure patterns: the original batch had early failures and a long-life tail, while the new batch had a more concentrated life distribution. In the two-parameter Weibull fits, the new batch had a higher B10 but a lower estimated mean life. These results do not establish an improvement in overall reliability.

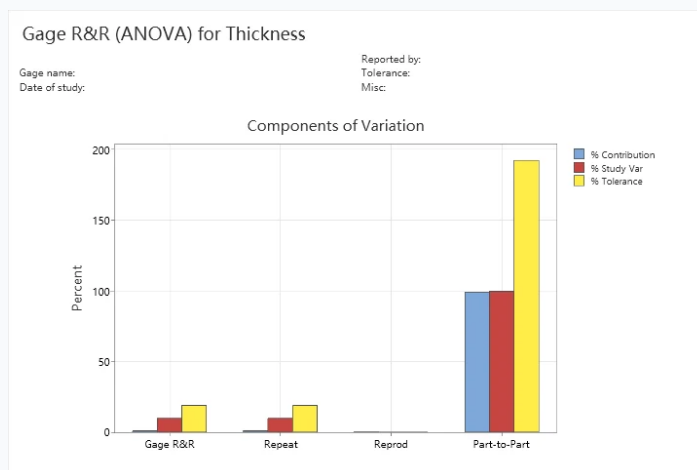
Fitted metric	Original batch	New batch	Interpretation
Weibull shape parameter β	Approx. 0.55	Approx. 4.06	Different patterns of failure risk over usage cycles
B10 life	Approx. 1.08 cycles	Approx. 17.9 cycles	Model-estimated point at 10% cumulative failure
Mean life (MTTF)	Approx. 110.3 cycles	Approx. 28.2 cycles	The original batch’s long tail raises its mean life



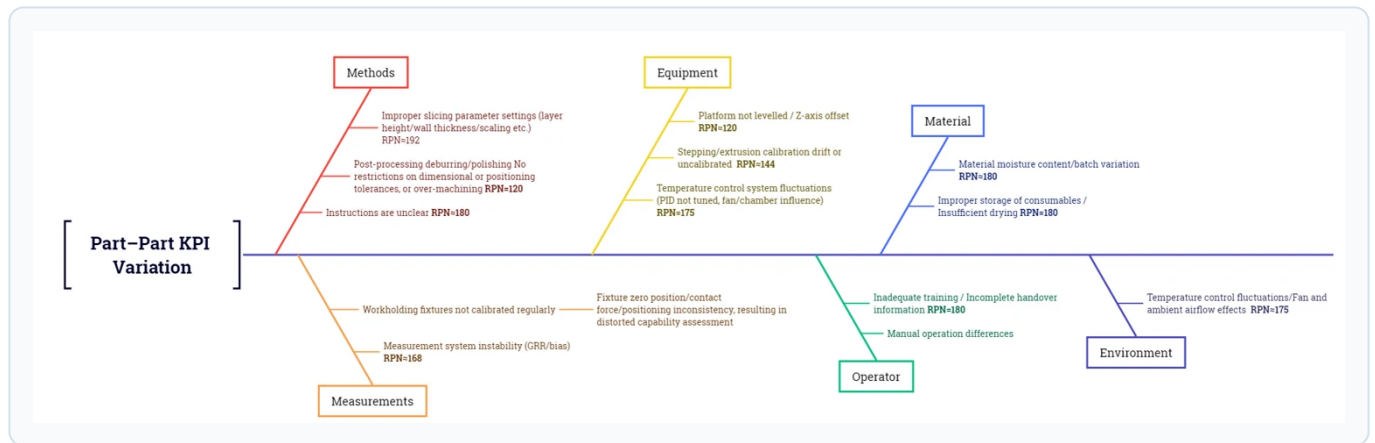
Weibull comparison of the original and new batches, original report, PDF p. 86.

Conclusions I Retain

- For dimensions that failed MSA, process capability results are exploratory only.
- Life distributions and means must be considered together; small-sample fits have substantial interval uncertainty.
- Improvements need to be assessed through dimensional centring, variation and bending life, rather than a narrower dimensional distribution alone.



Thickness Gage R&R variation breakdown, original report, PDF p. 11.



3D printing process fishbone diagram, original report, PDF p. 39.

Data and scope

%GRR is expressed as a percentage of study variation. Width and length failed MSA. Life results are fitted estimates from small samples of 10 parts per group and cannot establish reliability in volume production.

MSA: PDF pp. 10-21 ➤ Capability & FMEA: pp. 29-43 ➤ Life comparison: pp. 82-86 ➤

Beverage Production Planning, Bottlenecks & Inventory Simulation

Course group project AnyLogic · Excel · Discrete-Event Simulation · Demand Forecasting

Linked demand forecasting, production configuration, inventory and profit analysis across three beverage types and three package sizes to compare weekly production decisions.

My contribution

According to the report's contribution page, my main focus was demand forecast uncertainty and delays in production and supply chain responses. I proposed improvements to parameter adjustment, dynamic scheduling and data integration.

Project outputs

Analysis of demand forecasting and production configuration, plus findings on forecast uncertainty, response delays and model improvement recommendations.

9

Product-package combinations

30 weeks

Production decision framework

Filling

Main constraint identified by the team

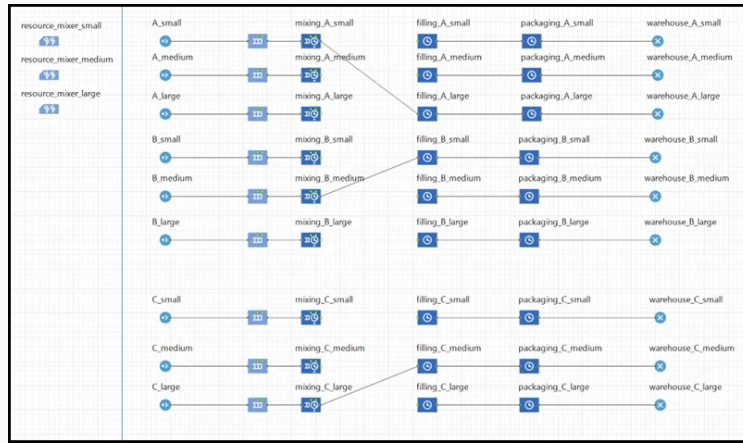
How the Team Made Decisions

The team used demand forecasts as inputs to simulate mixing, filling and packaging, then compared configurations using inventory by package size, revenue and holding costs. Each model agent represented 10 L of beverage, requiring consistent checks of conversions between bottle counts, litres and output units.

The team identified filling as the constraint on total output, so it prioritised comparisons of filling resources, product sizes and changeover strategies rather than increasing resources equally across all processes.

My Analysis Focus

Short forecast windows respond quickly but can trigger frequent changeovers. Longer windows are more stable but may lag behind demand changes. I analysed forecasting limitations, production response delays, delayed inventory feedback and external disturbances separately, then proposed ways to improve model responsiveness and data integration.



The team's AnyLogic production process model, original report, PDF p. 7.

Bottle type Type	Bottle Type	Demand	Product Cost	Production Cost	Warehouse Cost/product	Warehouse Cost	Bottle type Type	Total Cost	Product Price	Selling Price	Profit
A	330 ml	45000	0.9	40500	1	45000	A	40500	1.5	67500	27000
A	600 mL	28000	1.6	44800	2	56000	A	44800	3	84000	39200
A	1.25 L	13000	2	26000	3	39000	A	26000	4	52000	26000
B	330 ml	20000	0.7	14000	1	20000	B	14000	1.2	24000	10000
B	600 mL	24000	1.1	26400	2	48000	B	26400	2.3	55200	28800
B	1.25 L	8000	1.5	12000	3	24000	B	12000	3.5	28000	16000
C	330 ml	65000	1.1	71500	1	65000	C	71500	1.8	117000	45500
C	600 mL	34000	2	68000	2	68000	C	68000	3.5	119000	51000
C	1.25 L	13000	3	39000	3	39000	C	39000	4.5	58500	19500

Excel weekly product configuration and profit analysis table, original report, PDF p. 10.

PROCESS & ENGINEERING JUDGEMENT

Turning Forecasts, Capacity & Inventory into Weekly Plans

The team used 9 product-package combinations to develop a 30-week production decision framework. My main analysis covered forecast uncertainty, response delays and model improvements.

01 Compare forecasting methods and their suitability

The report compared moving averages, weighted moving averages and exponential smoothing, and used weighted moving averages to estimate demand. My analysis focused on the limitations of forecast windows and fixed parameters: short windows respond faster to recent changes but can prompt plan revisions; longer windows are steadier but may lag behind demand. Forecast errors need to be understood through their implications for inventory, shortages and changeovers.

02 Compare processes using consistent output units

Mixing, filling and packaging use different resources and processing methods. The team compared capacity across stages on a common basis and identified filling as the main constraint. Each model agent represented 10 L of beverage, making conversions between bottle counts, volume, time and weekly output particularly important. Increasing upstream output does not necessarily increase final deliveries and may instead create waiting or inventory buildup.

03 Connect production configuration with inventory feedback

The team used the AnyLogic model and Excel configuration tables to compare weekly production, inventory and profit. I focused on delays between demand forecasts, production responses and inventory feedback, proposing dynamic parameter adjustments, consideration of changeover effects and less manual data transfer. Not all 30 weeks were completed within the workshop time; the report includes later calculations. I therefore present the framework and review findings without claiming an overall profit improvement that has not been consistently verified.

Decision stage	Constraints to check together
Forecast inputs	Window length, weights, demand changes & forecast error
Production configuration	Filling constraint, package sizes, changeovers & time units
Weekly feedback	Inventory, shortages, holding costs & data transfer delays

Original report: ISS_A2_Group6_v3 / PDF pp. 3-14 and 20-22; individual contributions on p. 3.

Uncertainty	Impact on the plan	Recommendation in the report
Demand variation & fixed forecast parameters	Overproduction or shortages	Adjust forecast parameters as demand changes
Frequent production switches	Changeovers consume production time	Include changeover time and cost in scheduling
Delayed supply & inventory feedback	Inventory-demand mismatch	Review decisions against replenishment lead times and safety stock
Manual conversion between Excel & simulation	Input errors & decision delays	Standardise units and reduce manual data transfer

Scope & Review

The group did not complete all 30 weeks within the allotted workshop time, and the report records subsequent calculations. This portfolio therefore presents the decision framework, bottleneck analysis and improvement recommendations, without using overall profit or improvement rates that have not been consistently verified.

Dynamic parameter optimisation, intelligent scheduling and automated data interfaces are recommendations for future work. The existing report does not demonstrate that these functions were implemented.

Data and scope

The model and profit tables are group outputs; see PDF p. 3 for individual contributions. This case focuses on the production decision framework, constraint analysis and model review.

Individual contributions: [PDF p. 3](#) ➤ [Model & configuration: pp. 7-10](#) ➤ [Review & individual analysis: pp. 12-14](#) ➤

Five Manufacturing Processes and Dimensional Variation in Prusa Parts

Individual course report

Key Inspection Characteristics (KIC) · Tolerance checks · Process comparison · SolidWorks

Compared CNC machining, metal casting, PU casting, injection moulding and sheet metal fabrication for Prusa MK3.1 structural parts, examining dimensional and assembly issues when moving from prototypes to volume production.

My contribution

Prepared an individual report covering key dimension checks and nominal-to-measured comparisons for five types of parts, using physical samples and engineering drawings to compare process performance and propose improvements.

Project outputs

KIC dimension comparisons across five processes, manufacturing performance comparisons and sample defect analysis.

5

Manufacturing processes

KIC

Key inspection dimension comparisons

−0.29 mm

Injection-moulded guide-rod bore deviation from nominal

Linking dimensional variation to assembly risk

For example, the guide-rod bore in the injection-moulded Y-axis rod holder had a nominal diameter of 8.00 mm and a reported measurement of 7.71 mm: a deviation of −0.29 mm. Beyond a simple pass/fail judgement, the question is how this difference affects rod insertion, location and subsequent assembly.

Comparing process trade-offs

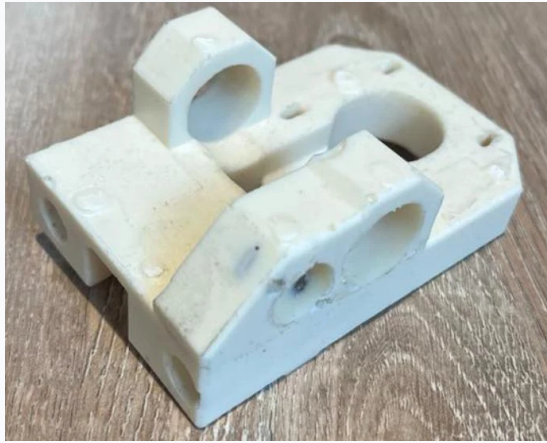
CNC machining calls for attention to workholding, tool access and machining accuracy; casting and injection moulding to shrinkage, mould design and cooling; and sheet metal fabrication to bends, hole positions and assembly interfaces. The report also compares speed, cost, quality, flexibility and sustainability.



Metal casting sample, PDF p. 11.



Sheet metal sample, PDF p. 21.



PU casting sample, PDF p. 14.

PROCESS & ENGINEERING JUDGEMENT

From critical dimensions to process trade-offs

Compared physical samples with engineering drawings across five manufacturing processes to understand how dimensional variation affects assembly, function and process selection.

01 Link critical dimensions to function

KIC checks covered bore diameter, hole spacing, clearance, position and perpendicularity. Differences between nominal and measured dimensions must be linked to specific functions, such as guide-rod insertion, bearing location, belt travel or motor mounting. For example, a nominal 8.00 mm injection-moulded guide-rod bore measured 7.71 mm; the first question is whether this affects rod insertion and location.

02 Compare the constraints of five processes

For CNC parts, the focus was on datums, workholding and tool access; for casting and injection moulding, on shrinkage, mould design and cooling. PU casting also required attention to mould alignment and deformation. Sheet metal fabrication required consideration of flat patterns, bending, springback and mounting-hole relationships together. The report compares speed, cost, quality, flexibility and sustainability to avoid selecting a process based on a single dimension.

03 Separate unusual readings from confirmed causes

The dimensional comparisons help locate problems, but a single measurement cannot establish a cause. Large deviations first require checks of the measurement datum, contact position and recording, followed by repeat measurements. Only after measurement effects are ruled out should process hypotheses, such as shrinkage compensation, workholding repeatability, cooling or springback, be investigated.

Inspection sequence	Questions to answer
Nominal and measured dimensions	What are the direction and magnitude of the deviation? Were measurements taken on a consistent basis?
Dimensions and assembly function	Does the deviation affect insertion, location, clearance or movement?

Inspection sequence	Questions to answer
Process and verification	Which factors might explain the deviation? How should it be remeasured?

Source report: Manufacturing Process and DFM Analysis / PDF pp. 4–22.

Process and part	Inspection focus	Potential improvements to verify
CNC / Z-axis top bracket	Bore diameter, hole spacing and perpendicularity	Review datums, workholding repeatability and toolpaths
Metal casting / Y-axis tensioner	Belt clearance, tensioning holes and mounting-hole positions	Review mould dimensions, shrinkage compensation and cooling
PU casting / X-axis motor bracket	Guide-rod bores, bearing bores and mounting positions	Check measurement datums, mould alignment and deformation
Injection moulding / Y-axis rod holder	Guide-rod bore diameter, bolt-hole spacing and clearance	Optimise mould design and cooling, then repeat measurements
Sheet metal / Z-axis motor bracket	Bend perpendicularity and mounting-hole relationships	Check flat patterns, bending springback and location

Engineering judgement

A single dimensional deviation can help locate a problem but cannot establish its cause. Large anomalous readings should be remeasured and measurement datums checked before process causes are investigated. Process selection must also consider production volume, part geometry and tooling investment, rather than relying solely on the general advantages and disadvantages of a process.

Data and scope

Deviations come from course sample records; proposed causes remain hypotheses requiring verification. This case concerns MK3.1 parts and is separate from the MK4/MK4S eight-station assembly data.

Study scope: PDF p. 4 ↗ Injection moulding measurements: pp. 15–17 ↗ Sheet metal and conclusions: pp. 18–22 ↗

Redesigning 3D-Printed Parts for CNC, Injection Moulding and Sheet Metal

Individual design report

SolidWorks · Engineering drawings · PrusaSlicer · Cost estimation

Adapted materials, holes and slots, wall thicknesses and bends to the manufacturing process while retaining assembly functions, using engineering drawings and estimates to explain design decisions.

My contribution

Prepared an individual design report covering physical 3D prints and slicer settings, together with manufacturing proposals, models and engineering drawings for the 401/402 brackets, 205Y tensioner bracket and 304X clamp.

Project outputs

Three manufacturing conversion proposals, part models, 2D engineering drawings and estimated cost breakdowns.

3

Manufacturing conversion proposals

2°-3°

Recommended injection-moulding draft angle

3D + 2D

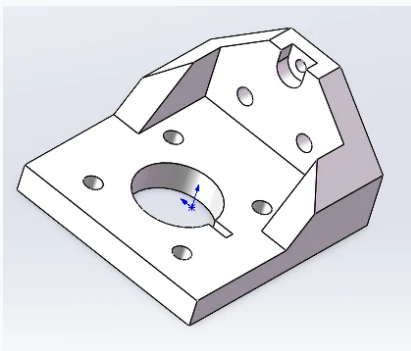
Models and engineering drawings

CNC brackets

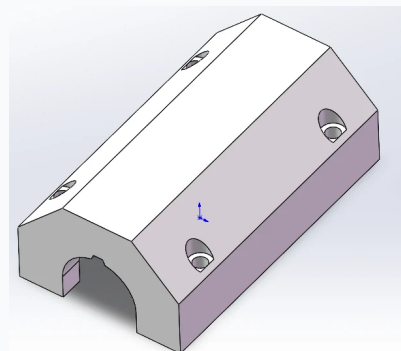
The L-shaped 401/402 brackets mount the Z-axis motor. The proposed design uses aluminium alloy, machinable solid sections and weight-reduction pockets, internal fillets instead of sharp corners, through-holes and simplified profiles to reduce enclosed cavities and complex toolpaths.

Injection moulding and sheet metal

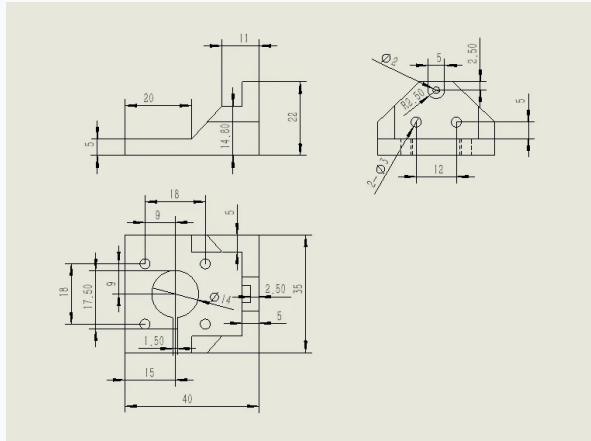
The 205Y injection-moulded bracket design considers draft, wall thickness, ribs, parting lines and shrinkage compensation. The 304X clamp is redesigned as a U-shaped stainless-steel bend that uses metal elasticity for clamping while retaining mounting holes and fit requirements.



CNC bracket model, PDF p. 2.



Sheet metal clamp model, PDF p. 8.



CNC bracket engineering drawing, PDF p. 3.

PROCESS & ENGINEERING JUDGEMENT

Adapting part geometry to CNC, injection moulding and sheet metal

Started with the assembly functions of existing printed parts, then adapted materials, geometry and manufacturing processes, using models, engineering drawings and estimates to explain the choices.

01 CNC brackets: allow for tooling and workholding

The 401/402 parts are symmetrical L-shaped brackets for mounting the Z-axis stepper motor. The proposal replaces the original printed infill structure with solid aluminium-alloy sections and weight-reduction pockets, using internal fillets, through-holes and simplified profiles to avoid enclosed cavities and complex toolpaths. The focus is on retaining the frame and motor mounting interfaces while making machining, location and inspection more practical.

02 Injection moulding and sheet metal: design around the forming process

The 205Y injection-moulded tensioner bracket design considers the mould-release direction, wall thickness, ribs, parting lines and shrinkage compensation, with a recommended draft angle of approximately 2° – 3° . The 304X support uses a U-shaped stainless-steel bend and metal elasticity to provide support, while retaining the required mounting holes and assembly relationships. Each process imposes different geometric constraints.

03 Connect design choices to cost and verification

The report presents the designs through SolidWorks models, 2D engineering drawings and PrusaSlicer print information, and estimates costs from materials, equipment hourly rates, processing time and amortised tooling costs. Cost estimates and simplified strength calculations mainly support comparisons between proposals; volume-production decisions also require defined production quantities, yields, workholding, actual loads and quotation terms.

Manufacturing conversion	Key design considerations
3D printing → CNC	Solid sections and weight-reduction pockets, internal fillets, through-holes and tool access

Manufacturing conversion	Key design considerations
3D printing → Injection moulding	Draft, uniform wall thickness, ribs, parting lines and shrinkage
3D printing → Sheet metal	Flat patterns and bends, material elasticity, hole positions and mounting interfaces

Source report: Part Design for Manufacture / PDF pp. 1–10; CNC proposal explanation on p. 4.

Part	Main design changes	Manufacturing and assembly considerations
401/402 motor brackets	Aluminium alloy, through-holes, fillets and weight-reduction pockets	Tool access, workholding, stiffness and location
205Y tensioner bracket	Recommended 2°–3° draft angle, 3–4 mm wall thickness and ribs	Mould release, sink marks, shrinkage and bearing-bore fit
304X clamp	1.0–1.5 mm stainless steel, U-shaped bend	Springback, clamping force, contact surfaces and bolt positions

Using the cost estimates

The report estimates unit cost from material mass, equipment hourly rates, processing time and amortised tooling costs. These figures help explain the cost components of each proposal; actual selection still requires a consistent basis for production volume, material quotations, mould life and yield.

Strength and safety-factor assessments require fully defined loads and constraints. This page retains the design rationale and verifiable geometric parameters, without presenting the original report's simplified strength estimates as conclusions that have passed engineering validation.

Data and scope

The figures show course design models. Manufacturing costs are estimates; the CNC, injection-moulding and sheet metal proposals have not been validated for volume production.

[CNC design: PDF pp. 2–5 ↗](#) [Injection moulding proposal: pp. 5–7 ↗](#) [Sheet metal proposal: pp. 8–10 ↗](#)

GPR and EM Loupe Mine-Site Investigation

Capstone project · Project leader

GPR · EM Loupe · EKKO Project · EmaxAIR · OriginPro

From site coordination and equipment preparation to mine-site data collection and integrated interpretation, used complementary survey methods to assess shallow structures and conductivity changes.

My contribution

As Capstone project leader, I coordinated with the mining organisation and progressed field-test arrangements at Costerfield Mine. I also contributed to equipment preparation, data collection and organisation, integrated interpretation, reporting and presentations.

Project outputs

Site coordination and test arrangements, survey-line data, radar profiles, conductivity maps and an integrated team report.

180 × 180 m

Costerfield study area

2 + 18

GPR profiles and usable EM survey lines

50 MHz

Mine-site GPR antenna frequency

Fieldwork

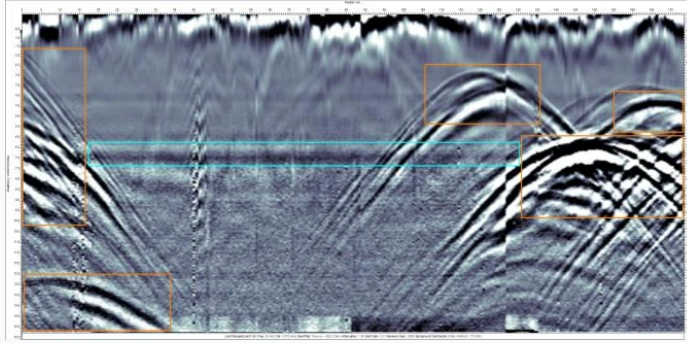
The team practised equipment use and data workflows at Royal Park before surveying Costerfield. Fieldwork required consideration of safety, access, weather, survey coverage and electromagnetic interference, while maintaining consistent survey-line directions and equipment geometry.

Data and interpretation

GPR provided shallow reflection profiles, while EM Loupe mapped conductivity over a greater depth range. The team organised the data, performed basic GPR processing and generated conductivity-depth images, combining both methods to develop a preliminary interpretation of subsurface layering.



Mine-site GPR data collection, team report PDF p. 17.



Interpreted mine-site GPR profile, PDF p. 30.

PROCESS & ENGINEERING JUDGEMENT

Field coordination and technical integration as project leader

As Capstone project leader, I coordinated with the mining organisation and arranged field testing, while contributing to data collection and organisation, integrated interpretation, reporting and presentations.

01 Project coordination: from practice site to mine site

The team first became familiar with the equipment, survey-line acquisition, recordkeeping and data checks at Royal Park, then conducted field surveys at Costerfield Mine. As project leader, I liaised with the mining organisation and progressed arrangements for test objectives, equipment and personnel access. Field coordination had to account for safety, access, weather, survey coverage and equipment coordination together, so that the data could be interpreted in its field context.

02 Technical integration: complementary evidence from two methods

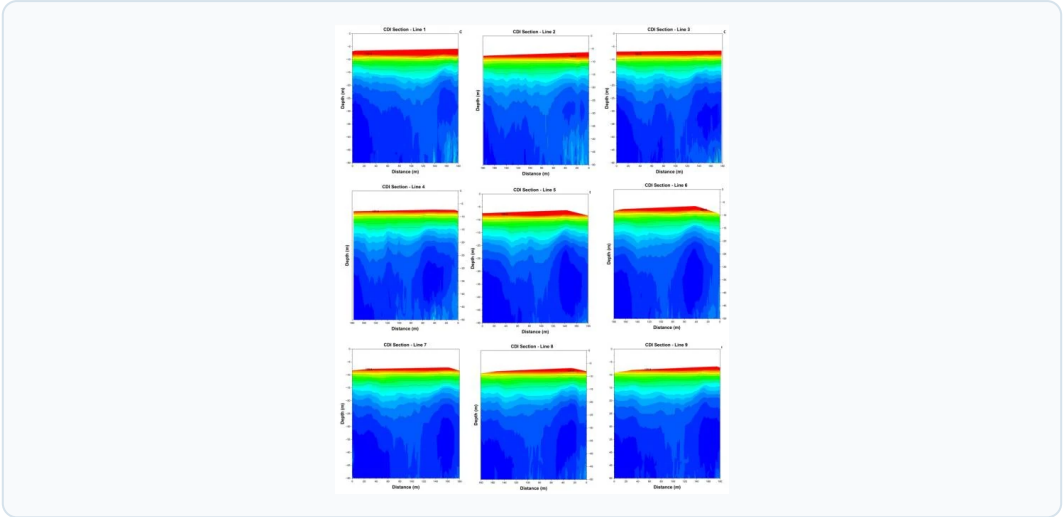
The team used EKKO Project to organise GPR data and perform time-zero correction, dewow filtering, background removal, gain adjustment and velocity fitting. EM Loupe data were processed using dsQC, EmaxAir and OriginPro for format conversion, inversion and conductivity visualisation. GPR characterised shallow reflection structures, while EM provided a broader conductivity context. Processing and interpretation were shared team outcomes.

03 Reporting and judgement: findings alongside limitations

The usable mine-site results comprised 2 GPR profiles and 18 EM survey lines. I contributed to testing, data organisation, preliminary interpretation, reporting and presentations, connecting field conditions and processed results back to the survey objectives. GPR interpretation focused on approximately the upper 10 m, and EM interpretation mainly on the upper 50 m. With no borehole or sampling verification along the survey lines, anomalies cannot be described as confirmed ore-body discoveries.

Responsibilities and outcomes	Details
My role: project leader	Mining organisation liaison, test coordination, field participation, reporting and presentations
Team: acquisition and processing	GPR profiles, EM survey lines, conductivity maps and integrated interpretation
Shared: further verification	Document coverage and data limitations; recommend boreholes, sampling or additional surveys for verification

Stage	Work undertaken	Outputs
Initial coordination	Discuss test objectives, candidate survey areas and site-access requirements	Field arrangements and survey scope
Equipment and acquisition	Check equipment settings, record along survey lines and back up data	GPR and EM field data
Processing and visualisation	Time-zero correction, dewow filtering, background removal, gain adjustment and CDI compilation	Radar profiles, conductivity-depth images and slices
Integrated interpretation	Compare shallow reflections with conductivity changes, identifying interference and limitations	Interpretation figures, technical report and presentations



Mine-site conductivity-depth section, source report PDF p. 32.

Engineering judgement and limitations

Rain and site conditions limited GPR acquisition to two boundary lines. Of the 19 planned EM survey lines, 1 was lost due to recording issues, leaving 18. The report limits reliable interpretation to approximately 50 m depth and notes sparse data at greater depths.

Without boreholes, sampling or trial pits along the survey lines, anomalies support only preliminary interpretations. Elevated conductivity may reflect water, clay, weathering or other factors and cannot be equated directly with mineralisation. Principal supervisor: Prof. Guillermo Narsilio.

Data and scope

The field survey and integrated interpretation were team outcomes. Anomaly interpretations have not been verified by boreholes or sampling along the survey lines and support preliminary investigation and planning of further tests.

Field plan: PDF pp. 15–19 ↗ Mine-site results: pp. 29–35 ↗ Limitations and conclusions: pp. 36–37 ↗

Wireless Mouse Life Cycle, Transport Scenarios and Ecodesign

Group coursework · Two reports

openLCA · ecoinvent · ReCiPe · LCC / NPV · Sensitivity analysis

Starting with product disassembly and a material inventory, the study brings raw materials, manufacturing, transport, use and end-of-life treatment within a common boundary to compare transport and product design options.

My contribution

I contributed to this life cycle project as a coursework team member. This page presents the team's material inventory, environmental impact analysis, cost framework and comparison of design options.

Project outputs

A material inventory, environmental contributions by life cycle stage, transport sensitivity analysis and a comparison of three ecodesign approaches.

3 years

Use period in the baseline functional unit

1.748 kg

Baseline model GWP, CO₂ eq

3

Alternative ecodesign approaches

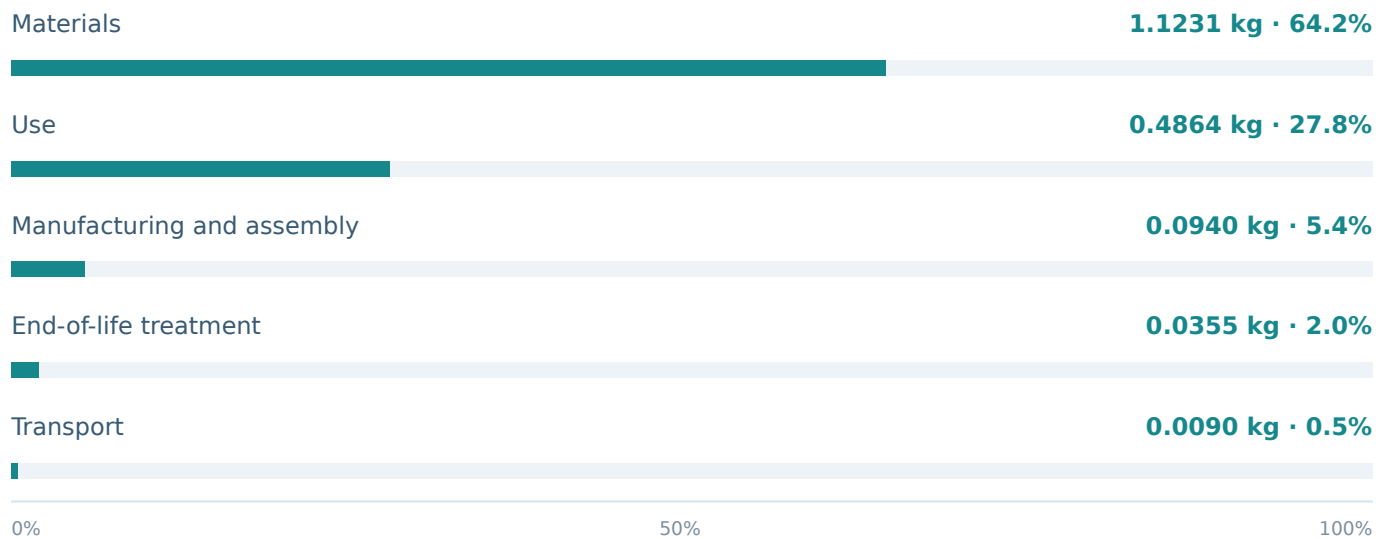
Set comparable boundaries

The baseline functional unit is one wireless mouse providing approximately 400 clicks per day over three years. Disassembly identifies the materials and their measured masses; manufacturing, logistics, battery use and waste treatment are then mapped to the life cycle inventory.

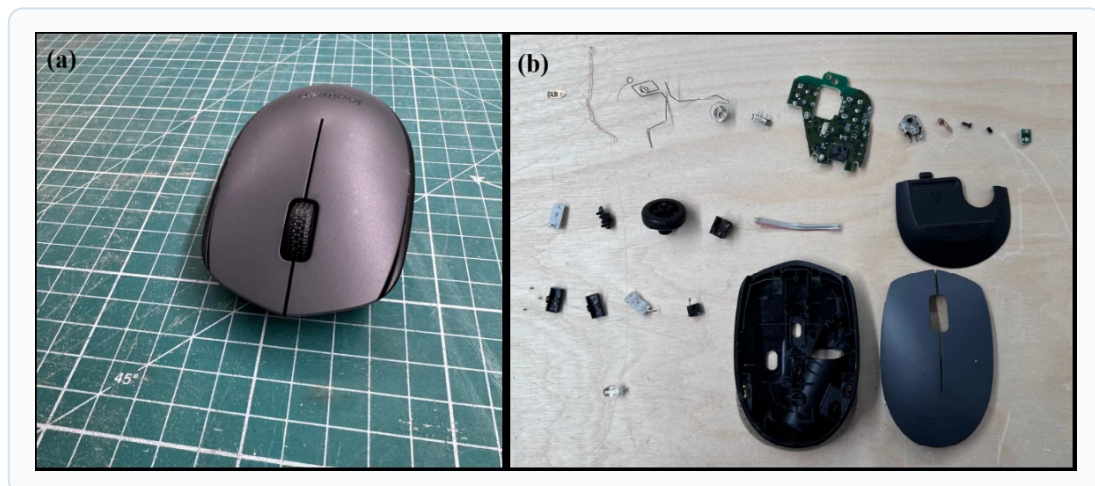
The global warming potential data in the SLCA appendix identify materials and use as the largest contributing stages, making the circuit board, plastics and power supply the main priorities for design discussion.

Life cycle GWP breakdown

Total: approximately 1.748 kg CO₂ eq · Three-year use period



Redrawn from the SLCA appendix; values in kg CO₂ eq. Percentages use the original values; displayed figures are rounded.



Wireless mouse before and after disassembly, with its components. SLCA report, PDF p. 5.



Exploded model view of the solar-powered mouse concept. LCC report, PDF p. 13.

Environmental hotspots, transport and cost within one boundary

Together, the two wireless mouse reports cover the baseline product, transport sensitivity and ecodesign. The analysis starts with the functional unit before examining environmental and cost trade-offs.

01 Build comparable systems from the disassembly inventory

The baseline functional unit is one wireless mouse providing approximately 400 clicks per day over three years. The team identified materials and components through disassembly, placed materials, manufacturing, transport, battery use and end-of-life treatment within the same system boundary, and compared environmental impacts using the ReCiPe 2016 midpoint method. Numerical comparisons between options are meaningful only when their function, use period and boundaries are consistent.

02 Identify hotspots, then compare transport scenarios

Baseline global warming potential is approximately 1.748 kg CO₂ eq, with materials contributing around 64.2% and use around 27.8%. Materials and the power supply are therefore the main areas for discussion. Transport accounts for approximately 0.51% of the total. Rail, road and other scenarios change the transport-stage result, but an improvement in transport cannot be treated as an equivalent percentage improvement across the mouse's entire life cycle.

03 Assess ecodesign alongside life cycle cost

The team compared three approaches: solar power, recycled ABS and a longer service life. The solar option also adds material and manufacturing burdens from photovoltaic components, so its net impact must be assessed over the full life cycle. The cost analysis uses LCC and discounting to examine expenditure at different stages. Because the original report's aggregation basis still requires checking, this portfolio retains the cost components and method without presenting unreconciled totals as a definitive quotation.

Comparison dimension	Key questions when interpreting results
Function and system boundary	Are use period, click demand, batteries and end-of-life treatment consistent?
Environment and transport	Does the improvement affect one stage or the full life cycle total?
Design and cost	Are added material burdens, lifetime assumptions and discounting treated consistently?

Source reports: SLCA, PDF pp. 4–12 and 18–19; Life Cycle Cost, Eco-Efficiency, PDF pp. 9–23.

Life cycle stage	GWP / kg CO ₂ eq	Approximate share
Materials	1.1231	64.2%
Use	0.4864	27.8%
Manufacturing and assembly	0.0940	5.4%

Life cycle stage	GWP / kg CO ₂ eq	Approximate share
End-of-life treatment	0.0355	2.0%
Transport	0.0090	0.5%

Transport scenarios and design options

SLCA compares rail transport in China, road transport in China and road transport in India. The LCC and ecodesign report further compares solar power, recycled ABS and a longer service life. Under the model's assumptions, the solar option performs better on most environmental indicators, but its power delivery and durability still require product testing.

Focus area	Main change	Conditions requiring further validation
Power supply	Use photovoltaics to reduce disposable battery consumption	Indoor light, energy storage and power reliability
Recycled materials	Reduce virgin ABS consumption	Material properties, availability and moulding conditions
Longer service life	Allocate material and manufacturing inputs over the service life	Actual durability, repairability and functional equivalence
Transport changes	Compare routes, transport modes and regional background data	Consistent mass-distance units, boundaries and databases

Cost and interpretation

The cost framework includes materials, manufacturing, transport, use and end-of-life treatment, using discounting to compare expenditure over the full life cycle. The original report's discounting table and some totals contain calculation issues that require checking, so this page does not present the total cost as a definitive quotation.

The transport share is recalculated from Appendix A.2 as approximately 0.51%, rather than the 0.05% stated in the main text. Environmental impact indicators have different units; this page explains them individually and does not add indicators with different dimensions into an overall score.

Data and scope

Environmental impacts are estimates from a coursework model. The material database, transport boundary, service life and power supply assumptions all affect the results. The study does not establish commercial product certification or demonstrated emissions reductions.

Disassembly and inventory: [SLCA pp. 4–7 ↗](#) Data and transport: [pp. 18–19 ↗](#) Cost: [LCC pp. 9–10 ↗](#) Ecodesign: [pp. 11–23 ↗](#)

↗

KFC Carlton Queueing and Service Resource Simulation

Coursework simulation report

AnyLogic · Customer behaviour · Queues · Scenario comparison

The model separates customer arrivals, ordering, waiting, collection and departure into process steps to examine how counter service, self-service ordering and kitchen resources jointly affect waiting times.

My contribution

The report presents an AnyLogic model, customer state logic, queue visualisations and improvement scenarios, focusing on how processes and resource allocation explain service congestion.

Project outputs

Models of ordering and collection, queue plots, and comparisons of service-counter and customer-routing scenarios.

2

Ordering routes

3 → 5

Service counters in the scenarios

Multiple queues

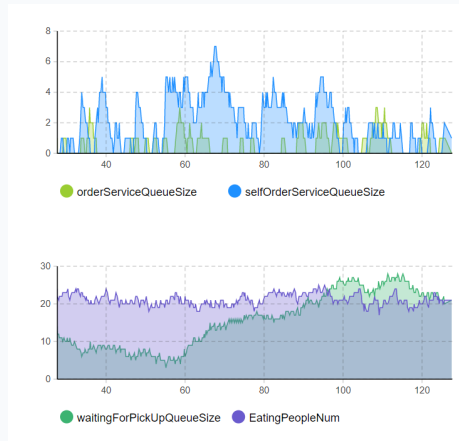
Ordering, collection and dining states

Identify constraints through the process

The model distinguishes counter ordering from electronic ordering and accounts for customer arrival rates, service times, seats, service counters and kitchen stations. It examines queue lengths in different states rather than using total restaurant occupancy alone to assess congestion.

Compare improvement scenarios

The report discusses additional counters, routing customers to self-service ordering and improving collection routes. The recorded scenarios include a simulated comparison of peak waiting time from 12 minutes to 8 minutes, illustrating the approach to comparing options.



Simulated ordering and collection queues and dining occupancy, from the original Word report.

PROCESS & ENGINEERING JUDGEMENT

Customer states, resource constraints and improvement scenarios

The service system is broken down into customer states and resource use to examine how ordering, collection and dining jointly create waiting.

01 Translate customer journeys into model logic

The report distinguishes counter ordering from electronic ordering and uses customer agents to represent entry, queueing, ordering, waiting, collection and departure. Service counters, kitchen stations and seats constrain the transitions between these states. This allows ordering queues and collection waits to be examined separately, avoiding an assessment of congestion based solely on the number of people inside the restaurant.

02 Compare resource allocation beyond adding staff

The report discusses increasing service counters from 3 to 5, routing customers to self-service ordering and improving collection routes. These scenarios require joint assessment of the counter and kitchen: a shorter ordering queue may shift the bottleneck to food preparation or collection. The 12-to-8-minute result is a scenario comparison from the report that illustrates the method; it is not an improvement demonstrated in an operating store.

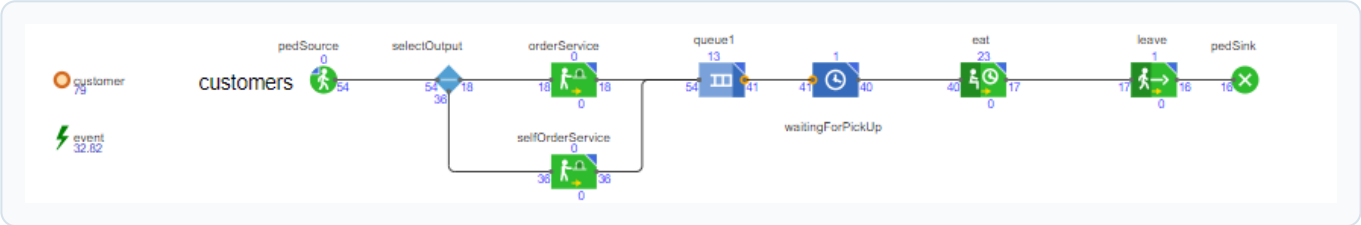
03 Define the next model-validation steps

Further validation should establish consistent peak and off-peak arrival inputs, service times and resource counts, and check whether customers can rejoin queues or leave early. Options should then be compared under identical input conditions, with repeated runs recording waiting times, queue lengths and resource utilisation. Where original input records and replicated experiments are insufficient, conclusions should remain within the scope of the coursework model.

Model component	What to observe
Customers and states	Arrivals, routing, queueing, waiting, collection and departure

Model component	What to observe
Service and resources	Service counters, kitchen stations, seats and collection routes
Scenarios and validation	Align inputs, compare scenarios and check for bottleneck shifts

Source report: Simulation and Optimisation Analysis of KFC Calton Queuing System / Word.



AnyLogic process model screenshot, from the original Word report.

Stage	Model observations	Scenario comparison focus
Ordering	Queue lengths at counters and self-service devices	Routing proportions and counter service capacity
Preparation and collection	Kitchen resource use and customers awaiting collection	Whether the bottleneck shifts after counter capacity increases
Space and departure	Seating and collection routes	Whether new congestion develops where flows intersect

How to use the results

The original report describes both peak arrivals of 120 customers/hour and model arrivals of 200 customers/hour, and lacks a rerunnable model and complete records of stochastic replications. The 12→8-minute comparison is therefore presented only as a reported scenario result, not a validated store improvement.

Further validation should reconcile arrival and service-time assumptions, define the interval over which waiting time is measured and run repeated experiments before comparing the combined effects of counters, staffing, kitchen resources and space constraints.

Data and scope

A coursework study of a service system, demonstrating model logic and resource-allocation scenarios. Simulation results require consistent inputs and repeated experiments before they can support operational decisions on site.

[Download original Word report ↗](#)

Toyota Supply Chain Strategy, Digitalisation and Resilience

Group coursework case study JIT / Kanban · MTS / MTO · EDI · Risk and sustainability

The study draws on cases and literature to examine production strategy, supplier collaboration, digital technology and risk response, analysing how information supports material flows and production decisions.

My contribution

As recorded in the team meeting notes, I was responsible for the introduction, analysis of Toyota's supply chain digitalisation, conclusion, and checks of the full report's structure and content.

Project outputs

Digital technology analysis, integration of the supply chain research, and checks of the full report's structure and content.

5

Research dimensions

Digitalisation

My main area of analysis

Case study

Based on literature and public sources

Research structure

The team examined five areas: production strategy, strategic partnerships, digitalisation, global supply chain risks and sustainability. The discussion covers pull systems and Kanban, combinations of MTS/MTO and supplier coordination, bringing inventory, lead times, responsiveness and risk into a shared framework.

My research contribution

I analysed how digital technologies support information synchronisation, visibility, forecasting and exception response, and linked this analysis to the report's other sections. I also checked consistency across the structure, introduction and conclusion.

How technology supports supply chain decisions

Technology area	Application in the analysis
EDI and information sharing	Coordination of orders, replenishment and delivery timing
IoT / RFID	Material status and traceability
Digital twins	Scenario assessment and risk response
AI and analytics tools	Demand forecasting and anomaly detection

Organised using the report's research framework to show the applications discussed.

PROCESS & ENGINEERING JUDGEMENT

Choose digital and collaborative methods around supply chain problems

This coursework study is based on public sources. I was responsible for the introduction, digitalisation analysis, conclusion, and checks of the full report's structure and coherence.

01 Organise supply chain research around a common framework

The team covered five areas: production strategy, strategic partnerships, digitalisation, global supply chain risks and sustainability, discussing pull systems and Kanban, MTS/MTO combinations and supplier coordination. When integrating the content, I checked that each section followed a consistent business logic: how demand feeds into planning, how materials support production, and how changes in information and supply affect inventory and lead times.

02 Connect digital technologies to specific business problems

My digitalisation analysis focused on information synchronisation, material visibility, forecasting and exception response. EDI relates to exchanging order and delivery information; IoT/RFID to material status and traceability; digital twins to scenarios and risks; and AI analytics tools to forecasting or anomaly detection. Naming a technology must be followed by an explanation of its use case, required data and the decisions it could inform.

03 Balance lean inventory, resilience and data quality

Lower inventory reduces capital tied up in stock but increases reliance on stable deliveries and timely information. The study therefore also considers supply disruptions, critical materials, inventory buffers and coordinated responses. I retained this trade-off in the conclusion: digitalisation proposals should first establish the problem and data quality, then consider tools. Insights from public case research should not be presented as systems I have implemented at Toyota.

Research output	Questions for further operational investigation
Order and supply coordination	Are changes synchronised promptly? Who handles shortages and delivery exceptions?
Data and visibility	Are data sources, update timing and traceability links reliable?
Risk and inventory	Are buffers and response plans aligned when critical material supply is disrupted?

Source report: TOYOTA'S SUPPLY CHAIN, PDF pp. 8–19; individual responsibilities on p. 28.

Research question	Key insight	Operational considerations
How should lean inventory be balanced against supply disruptions?	Low inventory requires reliable information, deliveries and risk responses	Lead-time variability, critical materials and safety stock
How should digital technologies be selected?	Match technologies to specific business problems	Whether data is timely, accurate and traceable
Why does coordination affect delivery?	Information and material movements must align across organisations	Order changes, arrival status and exception handling
How can sustainability inform decisions?	Consider materials, transport, suppliers and the use stage together	Trade-offs between cost, delivery and environmental objectives

Research scope

The report's technology recommendations are coursework research; I did not participate in Toyota's internal operations or systems implementation. The original report's technology scores of 1–10 are qualitative judgements. This page maps technologies to applications instead of presenting a numerical ranking.

Data and scope

A coursework case study based on public sources, focusing on digitalisation and supply chain decision analysis. The technology recommendations have not been implemented in an enterprise setting.

[Digital technologies: PDF pp. 8–12 ↗](#) [Risk and sustainability: pp. 13–19 ↗](#) [Individual responsibilities: p. 28 ↗](#)

PLC Conveyor Control & Robot Vision Sorting

Automation Coursework · Logic Development 2025.08-10 / 2026.09 Factory I/O · PLC · Niryo Studio · Blockly

Extended conveyor and detection control into a robot vision sorting workflow, translating ladder-logic sequence intent into colour routing, independent counters and quantity limits.

My contribution

Mapped inputs, outputs and action order; built branches and loops, and checked the connections between detection, picking, release commands and counter updates.

Current deliverable

Prior Factory I/O control simulation experience, plus a new Niryo two-colour Blockly sorting draft. Onsite parameter configuration and full physical validation remain pending.

2 + 2

Red / green target quantities

2

Independent counters and placement branches

4

Quantity guards across detection and picking

01 · Prior control simulation

Aug-Oct 2025: Worked on PLC I/O mapping and ladder-logic sequencing for Factory I/O conveyor, detection and storage scenarios, including start/stop, manual/automatic and interlock logic; checked stop and reset behaviour in simulation.

02 · Robot sorting extension

Sep 2026: Developed an observe-convey-detect-stop-pick-place workflow based on the course ladder sequence. Each colour has a target of two; completed colours are excluded from detection-triggered stops and further picks.

Engineering decisions

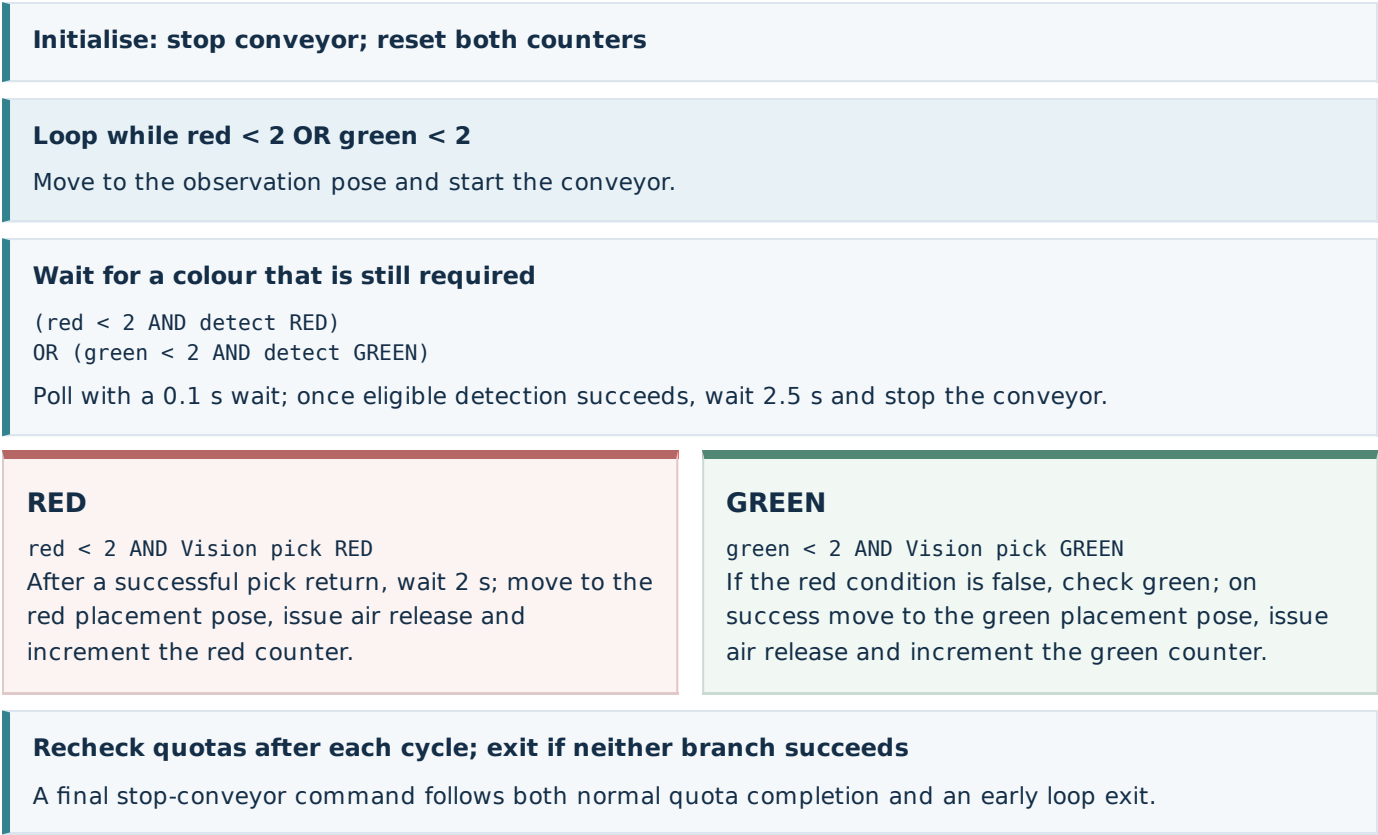
Control issue	Logic response
One colour reaches its target first	Use OR in the outer loop; guard each colour’s detection and pick with its own count < 2.
Placement and count ownership	Use separate placement branches and update only the corresponding counter after its release command.
Timing and unsuccessful return	Shared 2.5 s transport wait; an extra 2 s after red picking. If neither pick condition succeeds, exit the loop and execute the final conveyor stop.

Scope

The current output is a logic draft with block-by-block review. Target quantities are program settings, not measured physical picking results; the translation is not a scan-equivalent PLC implementation.

Combine “detected” with “still required”

The diagram summarises the constructed Blockly logic; it is not a screenshot of a running system.



Onsite validation plan

Configuration / check	What to observe
Poses and vision workspace	Configure observation and red/green placement poses and the calibrated workspace; check reachability and detection coverage.
Conveyor and picking parameters	Confirm speed, direction, pick-height offset and the onsite suitability of the 2.5 s and 2 s waits.
Quota and exit checks	Check red-first, green-first, alternating arrivals and unsuccessful pick returns; verify no excess picks, correct counters and a stop after exit.

Validation status

Full physical execution has not been confirmed. A pick return and completed release command are not independent grasp/drop sensor feedback; the exit logic is not a safety-rated emergency stop.

Evidence: course PLC ladder sheet, personal Blockly construction screenshots and prior Factory I/O project records.
Updated 21 Sep 2026.

Project setup & Blockly program

PLC Conveyor Control & Robot Vision Sorting · Project images

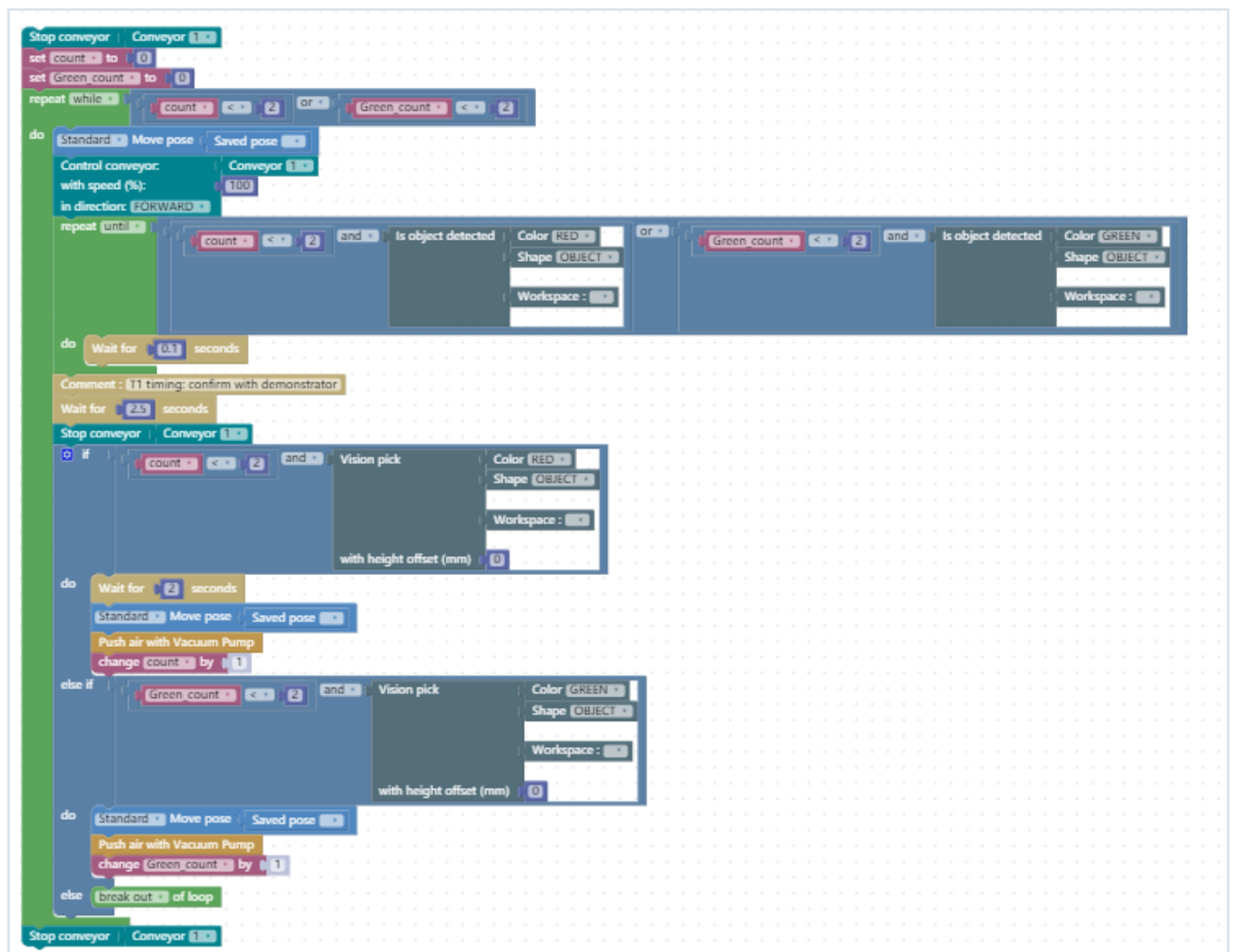


Niryo Ned robot and conveyor

The setup photograph shows the robot, end tool and conveyor. Red and green objects illustrate the colour-sorting scenario. The sequence covers observation, feeding, detection, picking and separate placement.

The images document the setup and logic structure. Saved poses and workspaces remain unselected in this screenshot; complete hardware-cycle validation is still pending.

Blockly draft: independent counters, two objects per colour



Original screenshot: quota guards, a red-only delay and counting after the release command.

Source: author-provided setup photograph and Niryo Studio program screenshot.

Education & experience

Connecting structural, data and process analysis from coursework with coordination and execution in practice.

EDUCATION

University of Melbourne · Master of Industrial Engineering

Jul 2024–Present · Candidate

Projects in industrial systems simulation, manufacturing and assembly, quality and reliability, life cycle assessment, supply chain management and Capstone engineering practice.

University of Melbourne · Bachelor of Science

2020.02–2022.12 · Mechatronics Systems

Specialised in Mechatronics Systems, building a foundation in mechanical structures, automation and systems analysis.

WORK EXPERIENCE

TST Chadstone Pty Ltd

Dec 2024–Present · Warehouse & Supply Chain Operations Assistant (part-time)

Support receiving, dispatch, inventory organisation, replenishment, order processing and logistics coordination; follow up on shortages, delivery delays and order discrepancies.

Risunny Technology Ltd

Nov 2020–Feb 2021 · Mechanical Engineering Intern

Supported hydraulic solenoid directional-valve design and manufacturing through SolidWorks modelling, engineering drawings and basic finite element analysis; assisted with production feedback and dispatch.

Methods & tools in practice

Focus area	Methods and tools used or explored	Supporting projects
Industrial engineering	Time studies, DFA, line balancing and standard work	Prusa assembly
Design for manufacture	SolidWorks, engineering drawings, PrusaSlicer and process-feasibility analysis	Five-process analysis, part design and internship
Quality & reliability	Minitab, Gage R&R, Cpk/Ppk, FMEA, Weibull	Hinge quality and life-data analysis
Systems & operations	AnyLogic, Excel, demand forecasting, queue and inventory analysis	Beverage simulation, KFC and TST experience
Life cycle	openLCA, material inventories, environmental hotspots, LCC and sensitivity analysis	Two wireless-mouse reports
Fieldwork & coordination	Equipment preparation, field data collection, technical communication, data organisation and reporting	GPR / EM Loupe Capstone

Nine source reports

All nine source reports are available to open or download, with links to the relevant figures and sections.

PDF

Probability, Reliability and Quality

PDF · 86 pages · 3.18 MB · Case 02

Probability, Reliability and Quality(1).pdf

[Open report ↗](#)

[Download](#)

PDF

Beverage Production Planning & Simulation

PDF · 22 pages · 0.74 MB · Case 03

ISS_A2_Group6_v3(1).pdf

[Open report ↗](#)

[Download](#)

PDF

Manufacturing Process and DFM Analysis

PDF · 24 pages · 1.34 MB · Case 04

Manufacturing Process and DFM Analysis.pdf

[Open report ↗](#)

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PDF

Part design for manufacture

PDF · 10 pages · 0.58 MB · Case 05

Part Design for Manufacture(5).pdf

[Open report ↗](#)

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PDF

GPR & EM Loupe Capstone

PDF · 43 pages · 2.84 MB · Case 06

Capstone Report 2026(5).pdf

[Open report ↗](#)

[Download](#)

PDF

Wireless Mouse: Environmental Life Cycle Assessment

PDF · 19 pages · 1.98 MB · Case 07

SLCA(3).pdf

[Open report ↗](#)

[Download](#)

PDF

Wireless Mouse: Cost & Ecodesign

PDF · 28 pages · 2.47 MB · Case 07

Life Cycle Cost, Eco-Efficiency(4).pdf

[Open report ↗](#)

[Download](#)

DOCX

KFC Queuing-System Simulation

Word source report · 0.11 MB · Case 08

Simulation and Optimisation Analysis of KFC Calton Queuing System(1).docx

[Download](#)

PDF

Toyota supply-chain study

PDF · 30 pages · 0.42 MB · Case 09

TOYOTA'S SUPPLY CHAIN(3).pdf

[Open report ↗](#)

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Source reports can be opened or downloaded from this website. The printed portfolio includes the case text and figures; download source attachments separately.

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Career interests: industrial engineering, quality, manufacturing, supply chain and project engineering

Shanghai preferred · Happy to discuss project methods and source materials

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